



Kratu

An offline, voice-first tutor that teaches non-literate Hausa-speaking children to read and speak English.

Kratu turns a low-cost tablet — with no internet required after setup — into a patient, one-to-one teacher. A child who cannot yet read taps a friendly guide, hears a word, sees a photograph from their own world, says it aloud, and is understood. Every lesson runs fully on-device, in the child's mother tongue, guided by voice so that literacy is never a prerequisite for learning.

DOCUMENT	DATE	PREPARED BY	STATUS
PRD v1.0	[Month YYYY]	[Applicant / Org]	Working prototype

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01 Executive Summary

What Kratu is, who it serves, and why it deserves funding — in one page.

Kratu is a self-contained learning application that teaches foundational English literacy — letters, sounds, vocabulary, spelling, and reading — to children who speak Hausa and cannot yet read in any language. It runs entirely offline on an ordinary tablet, is operated by voice and touch rather than text, and grounds every lesson in photographs and words drawn from the child's own environment in northern Nigeria.

The learning experience is anchored by **Laila**, a friendly on-screen guide. A child hears an instruction in Hausa, taps Laila, speaks a word in English, and Kratu — using on-device speech recognition — tells them instantly whether they were understood. No teacher needs to be literate in English; no lesson needs a network connection; no child needs to be able to read a single word to begin.

100%

Offline after install — no connectivity required to teach

0

Reading ability assumed of the child at start

130+

Photorealistic, context-relevant vocabulary images

8

Learning modes from alphabet to reading

A working prototype exists today: a single-file web application with staged A–Z alphabet lessons, thirteen vocabulary domains, spelling and reading modes, gender-aware voice guidance, and fully on-device speech recognition. **This grant funds the path from prototype to a field-ready learning programme** — content depth, a hardened offline installer, a facilitator toolkit, a supervised pilot at a physical learning centre, and an independent evaluation of learning gains.

SCOPE OF THIS REQUEST

Funding covers curriculum completion, a field-ready offline build and facilitator toolkit, a monitored pilot with [N] children, and an independent evaluation of learning gains — producing a template that can be replicated across underserved, low-connectivity communities.

02 The Problem & Context

A literacy emergency, concentrated where infrastructure is thinnest and mother-tongue tools are rarest.

Nigeria carries one of the largest out-of-school and low-literacy populations in the world, and that burden is not spread evenly. It falls hardest on the north, on rural communities, and on girls — precisely the children for whom conventional, text-first, connected, English-medium edtech is least usable.

~10–
18M[‡]

Out-of-school children in Nigeria — among the highest counts globally

50M+[‡]

First-language Hausa speakers across West Africa

North[‡]

Region where out-of-school and low-literacy rates concentrate

Girls[‡]

Disproportionately excluded from foundational literacy

Why existing tools do not reach these children

- **They assume literacy to teach literacy.** Most learning apps require the child to read menus, buttons, and instructions — a barrier for the very children who most need to learn to read.
- **They assume connectivity.** Streamed video, cloud accounts, and online speech services fail in communities with intermittent or absent internet and unreliable power.

- **They assume English as the language of instruction**, leaving a Hausa-speaking child with no bridge from what they know to what they are learning.
- **They assume an adult teacher fluent in English** is present and available for one-to-one practice — a scarce resource in under-resourced classrooms.
- **They are culturally distant** — illustrations, names, and examples drawn from contexts the child has never seen.

THE GAP KRATU FILLS

A learning tool that needs no reading ability, no internet, no English-literate adult, and no cultural translation — because it is voice-first, offline-first, mother-tongue-scaffolded, and built from the child's own world.

† Figures are indicative and must be replaced with current, cited statistics before submission. Suggested sources listed in §22.

03 Vision, Mission & Objectives

Vision

A generation of children in low-connectivity, mother-tongue communities who cross the threshold into literacy — regardless of whether a school, a network, or an English-speaking teacher can reach them.

Mission

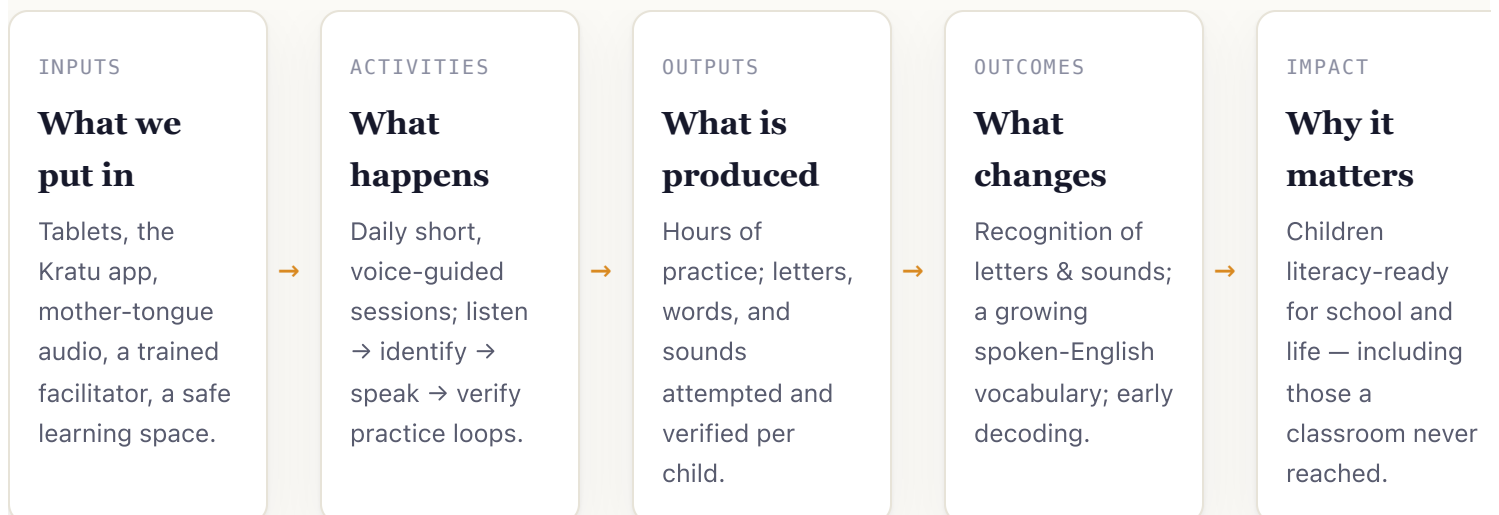
To put a patient, one-to-one, voice-guided literacy tutor into the hands of every child who has been left behind by text-first, connected, English-only education.

Programme objectives

#	OBJECTIVE	SUCCESS LOOKS LIKE
O1	Complete a coherent, staged curriculum from alphabet to first reading.	A child can progress end-to-end with no external instruction.
O2	Prove measurable learning gains in a supervised pilot.	Statistically meaningful pre/post improvement in letter, word, and sound recognition.
O3	Make deployment turnkey for a non-specialist facilitator.	A centre can install and run Kratu with a one-page guide.
O4	Guarantee it works with zero connectivity and modest hardware.	Full function on an entry-level tablet, offline, on battery/solar.
O5	Produce a replicable template for other languages and regions.	Content and voice layers swappable without re-engineering.

04 Theory of Change

How inputs become lasting literacy outcomes.



Core assumption. When the barriers of reading-ability, connectivity, language, and teacher-availability are removed, a motivated child with regular access to patient, responsive, mother-tongue practice will make foundational literacy gains. Kratu's design removes each of those four barriers by construction; the pilot (§15) exists to test the assumption and quantify the gain.

05 Beneficiaries & Personas

Designed around the people who will actually hold the device.



PRIMARY LEARNER

Amina, 7 — speaks Hausa, cannot yet read

Has never attended, or has irregularly attended, formal school. Curious and capable, but locked out of every tool that requires reading to begin. **Needs:** to be spoken to, not read at; instant, encouraging feedback; and pictures of a world she recognizes. **Kratu gives her:** a guide who talks, a microphone that understands her, and success on the very first tap.



FACILITATOR

Musa — community centre volunteer

Committed and trusted, but not a trained teacher and not fluent in English. **Needs:** a tool that teaches without him having to; simple setup; a way to see who is progressing. **Kratu gives him:** a self-driving lesson, a one-page install, and per-child progress at a glance.



HOST SITE

The learning centre — community, NGO, or faith-based

A room, a handful of tablets, intermittent power, and often no reliable internet (connectivity, where present, may be via satellite such as Starlink). **Needs:** tools that survive offline, share devices across many children, and require minimal supervision. **Kratu gives it:** a fully offline programme, per-child profiles on shared hardware, and low operating overhead.

06 Product Overview

Kratu presents as a warm, calm, full-screen application. A child chooses their profile, and is then guided entirely by voice through a set of learning modes that build on one another. Every screen has a consistent grammar: **a friendly guide who speaks, a large picture, and one or two large touch targets** — never a wall of text or a cluttered menu.

Guided by Laila

A 3-D animated mascot who gives instructions, invites the child to speak, and celebrates success — the single, consistent point of interaction.

Spoken, not written

All instruction is audio in Hausa; the target language (English) is heard and spoken. Reading is the goal, never the entry requirement.

Answered by voice

The child speaks into the device and is understood on-device — turning passive tapping into active production.

DESIGN NORTH STAR

A four-year-old who has never seen a screen should be able to succeed at their first lesson without a single word of written instruction.

07 Learning Model & Pedagogy

A tight, repeatable practice loop, staged so difficulty rises only as mastery grows.

The core practice loop

1 · Hear

Listen

Kratu names the item in Hausa, then models the English word.

2 · See

Anchor

A photograph from the child's world fixes meaning to the sound.

3 · Choose

Decide

"Do you know it in English?" — tap Laila to try, or the board to be taught.

4 · Speak

Produce

The child says the word aloud; the device listens on-device.

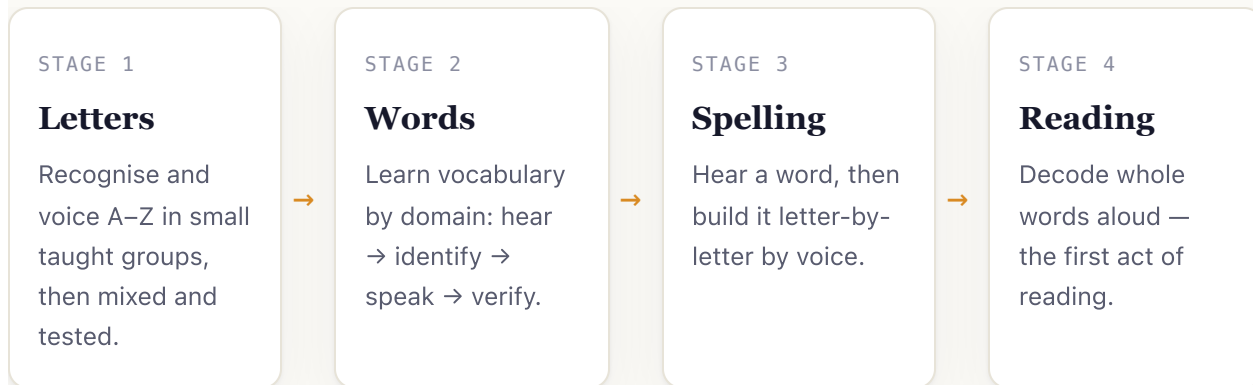
Verify

Instant,
encouraging
feedback —
correct, or gently
modelled again.

Pedagogical foundations

- **Mother-tongue scaffolding.** New English is always bridged from Hausa the child already owns — reducing cognitive load and honouring their first language.
- **Active production over passive recognition.** The child *speaks*; spoken production, verified in the moment, builds durable phonological memory that tap-only quizzes cannot.
- **Immediate, judgement-free feedback.** Every attempt is met at once — celebrated or re-modelled — never punished. Repetition is safe.
- **Staged mastery.** Content is chunked and gated: the alphabet is taught in small groups; vocabulary is learned then mixed then tested; each stage unlocks the next.
- **Dual coding.** Sound + image + spoken production together encode each item through multiple channels.
- **Low-stimulation calm.** No timers, scores, or loud game mechanics that overwhelm a first-time learner — attention stays on the word.

Progression



08 Feature Set

The learning modes and system features, with current build status.

Learning modes

MODE (HAUSA)	WHAT THE CHILD DOES	STATUS
Haruffa Alphabet	Learn and voice letters A–Z in staged groups, then mixed practice and a full test.	Built
Abubuwa Objects	Identify and speak everyday things across many domains, verified by voice.	Built
Rubutu Spelling	Hear a word and assemble it letter-by-letter, spoken aloud.	Built
Karatu Reading	Read whole words aloud — verified decoding.	Built
Launuka · Siffofi · Lambobi Colours · Shapes · Numbers	Foundational concept vocabulary as dedicated domains.	Built
Nemo hoto Listen & find	Hear a word and choose the matching picture — receptive practice.	Expanding

System features

Laila, the voice guide

A 3-D animated mascot — she blinks, breathes, and glows when inviting the child to speak. Pressing Laila directly turns on the listener; she *is* the microphone.

Gender-aware voice

Instructions are grammatically matched to the child's gender in Hausa (e.g. *ka/ki*), so guidance sounds natural and personal.

On-device speech recognition

The child's speech is recognised locally, with no audio ever leaving the device — the engine that makes "speak and be understood" possible offline.

Per-child progress

Named profiles track mastery per mode and domain on shared hardware, so many children can share one tablet.

Culturally grounded images

A library of photorealistic pictures shows animals, food, homes, and people the child actually knows — a goat, not a moose.

Fully offline

Every asset — audio, images, recogniser — ships inside the app. After install, nothing needs a network.

09 Content & Curriculum

The vocabulary domains, and the media that bring each word to life.

Vocabulary domains

Words are organised into recognisable domains so learning is contextual and cumulative:

Dabbobi · Animals

Abinci · Food

Kayan gida · Home

Jiki · Body

Yanayi · Nature

Motoci · Vehicles

Mutane · People

Ayyuka · Actions

Tufafi · Clothing

Makaranta · School

Launuka · Colours

Siffofi · Shapes

Lambobi · Numbers

13

Vocabulary domains, alphabet, spelling & reading

130+

Unique photorealistic images, culturally specific

2

Audio languages — Hausa guidance, English target

A–Z

Full alphabet with per-letter sound & word

Audio

Every instruction, word, and letter is a pre-generated voice clip in Hausa and English, bundled in the app — clear, consistent, and available offline.

Imagery

Photorealistic images generated and curated for cultural fit — the animals, foods, and people of the child's environment, not stock clip-art.

Grant-funded depth

This grant expands each domain toward richer coverage and adds early sentence and phrase work beyond single words.

EDITORIAL CONTROL

All content is reviewed by native Hausa speakers and educators for linguistic accuracy, cultural appropriateness, and pronunciation — a standing quality gate, not a one-off pass.

10 Experience & Accessibility

Principles that make the product usable by a non-literate first-time user.

No reading to learn to read

Every action is reachable by listening and tapping. Written labels are decorative reinforcement, never the only path.

One decision per screen

Large targets, generous spacing, and at most a couple of choices keep a first-time learner from ever being lost.

Consistent grammar

The same guide, in the same place, doing the same thing on every screen — predictability lowers anxiety and builds independence.

Forgiving & patient

No penalties, timers, or dead-ends. A child can replay, retry, or step back at any moment, endlessly.

Low-power, low-stimulation

Calm colour, minimal motion, no jarring sound — kinder on attention and on battery.

Robust on modest hardware

Designed for entry-level tablets and small screens, with touch targets sized for small hands.

11 Technology & Engineering

A reference description of how Kratu is built — the stack, the pipelines, the data model, and the on-device speech engine that makes an offline talking tutor possible.

Web

Open standards (HTML / CSS / JS) — no proprietary runtime

On-device

Speech recognition runs locally in WebAssembly

0

Network calls required to deliver a lesson

1 file

Self-contained application artifact

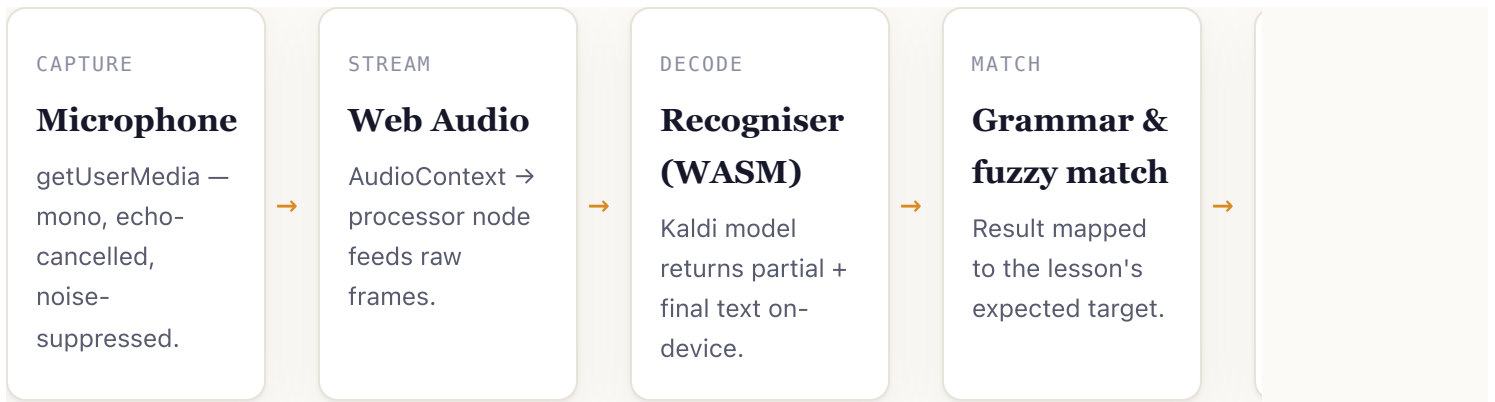
11.1 • System design at a glance

Experience	Full-screen web UI	Voice + touch, no reading	Animated SVG mascot (Laila)
Learning logic	Staged lesson engine	Mastery gating	Per-child progress store
Speech in	Vosk / Kaldi recogniser (WASM)	Grammar-constrained decoding	Web Audio capture pipeline
Speech out	Pre-generated Hausa + English clips	Gender-aware selection	Bundled, no network
Media	WebP image library	Encoded audio (data URIs)	Vector mascot & icons
Delivery	Single self-contained artifact	Cache-first / installable (roadmap)	Runs from local storage
Hardware	Entry-level Android tablet	Battery / solar	Optional satellite for updates only

Kratu is a **framework-free, single-artifact web application**. There is no server, no account system, and no cloud dependency in the learning path: the interface, lesson logic, speech recogniser, audio, and images all live inside the package and execute on the device. Being built on the open web platform means it runs across the widest range of low-cost hardware without per-platform rebuilds.

11.2 • On-device speech recognition

The enabling technology — and the hardest engineering — is recognising a child's speech **entirely on the device, with no connectivity**. Kratu uses a **Vosk** speech recogniser (built on the Kaldi engine) compiled to **WebAssembly**, with a compact acoustic model bundled in the app. Audio is captured through the browser's Web Audio pipeline and streamed to the recogniser frame-by-frame; nothing is ever transmitted off the device.



COMPONENT	DETAIL
Engine	Vosk (Kaldi-based) offline recogniser, compiled to WebAssembly for the browser.
Model	Compact bundled acoustic model ($\approx$ 40 MB), loaded once from local storage.
Audio capture	Web Audio API — single-channel, echo cancellation and noise suppression on; frames streamed live to the recogniser.
Grammar constraint	The recogniser is restricted to the <i>expected</i> tokens for the current task (the letters in a group, or a domain's words). This sharply raises accuracy on child speech and cuts compute.
Matching layer	Recognised text is mapped to the target with tolerant matching; up to a few forgiving retries before the word is re-modelled.
Privacy	All processing is local. No audio is recorded to disk, transmitted, or sent to any service.

DESIGN CONSEQUENCE

Because recognition is grammar-constrained per lesson, Kratu can be both accurate and fast on modest hardware — it only ever has to decide between a small, known set of expected answers, not transcribe open speech.

11.3 • Guidance audio (speech out)

Every instruction, letter, and word the child hears is a **pre-generated voice clip**, bundled inside the app as encoded audio so playback never touches the network. Two languages run in parallel: **Hausa** for all guidance, and **English** as the target being taught. Clips are selected at runtime, including **gender-aware variants** so instructions are grammatically correct for the child (Hausa distinguishes, e.g., *ka / ki*).

Current build

Guidance audio is produced from text-to-speech, layered and packaged by a build step, and embedded directly in the app as encoded clips.

Production path

For final quality and clean licensing, the programme moves guidance audio to **native-speaker recordings** (a grant content deliverable, §9/§19), keeping the same runtime.

11.4 • Visual content pipeline

Kratu's pictures are **photorealistic and culturally specific** — a goat, not a moose; local food, homes, and people. They are produced with modern **text-to-image diffusion models** (FLUX family) run locally, from a curated prompt library, then reviewed for cultural fit and pronunciation alignment.

STAGE	DETAIL
Generation	FLUX diffusion models (a higher-fidelity model for finals, a fast model for iteration) run locally on Apple-silicon hardware, driven by a versioned prompt library.
Curation	Every image is reviewed by native speakers/educators for accuracy and appropriateness before it enters the library.
Optimisation	Downscaled to ≈ 640 px on the long edge and re-encoded to WebP (~q90) , alpha-safe and visually lossless — small files for constrained storage.
Packaging	Optimised images are embedded in the app so they load instantly and fully offline.

11.5 • Lesson engine & data model

Content is data, not code. Vocabulary domains, the alphabet, and each mode's sequence are described declaratively; the same engine drives every mode. This is what makes new languages and regions a content exercise rather than a rebuild.

- **Content model.** Each domain is a list of items; an item carries its written word, its image, and the keys of its Hausa and English audio clips. The alphabet is grouped into small taught sets.
- **Phase machine.** A mode advances through phases — for words: *learn* → *mixed* → *test*; the alphabet: *taught group* → *mixed* → *full test*; plus dedicated spelling, reading, and listen-and-find phases.
- **Mastery gating.** Progress unlocks the next chunk; verified spoken answers, not mere taps, move the child forward.
- **Progress store.** Per-child profiles and mastery are persisted locally on the device (browser storage), keyed by profile — so many children share one tablet with independent progress and no personal data collected.

11.6 • Offline delivery, storage & footprint

Kratu ships as a **self-contained artifact** with all media inlined, so it runs from local storage with no network. Installable offline caching (service worker / PWA) so it launches like a native app from the home screen is a near-term **hardening deliverable** (§14, Phase 2).

CHARACTERISTIC	VALUE (INDICATIVE – CONFIRM ON SHIPPING BUILD)
Application + media bundle	≈ [15–25] MB (HTML, encoded audio, WebP images)
Speech model	≈ 40 MB (bundled once)
Total on-device footprint	≈ [~55–65] MB
Runtime dependencies	A modern mobile browser engine (bundled on any current Android tablet)
Connectivity	None to teach; optional network only to install or update
Distribution	Side-load once from a file or memory card; no app-store dependency required

11.7 • Interface, rendering & the mascot

The interface is standard web UI, tuned for a first-time, non-literate user: large targets, one decision per screen, calm colour, and reduced-motion awareness. **Laila** — the guide the child taps to speak — is a single **vector (SVG) mascot** with gradient 3-D shading and lightweight CSS animation (an idle breathe and a blink). Being vector, she is razor-sharp at any screen size and costs almost nothing in storage or battery. Pressing Laila directly switches on the listener: she *is* the microphone, so there is never a separate control to hunt for.

11.8 • Technology stack

LAYER	TECHNOLOGY	ROLE
Runtime	Open web platform (HTML, CSS, vanilla JS)	Cross-device UI & logic with no framework or build runtime
Speech-in	Vosk / Kaldi, WebAssembly	Offline, on-device speech recognition
Audio-in	Web Audio API + getUserMedia	Live microphone capture & framing
Speech-out	Pre-generated / recorded voice clips	Bundled Hausa & English guidance audio
Imagery	FLUX diffusion (local) → WebP	Culturally-specific vocabulary pictures
State	Browser local storage	Per-child profiles & progress, on-device
Graphics	Inline SVG + CSS	Scalable mascot, icons, animation
Delivery	Single artifact; PWA caching (roadmap)	Offline install & launch

11.9 • Engineering & content tooling

Behind the app sits a small toolchain that keeps content reproducible — so expanding the curriculum is a repeatable process, not bespoke work each time:

Audio toolchain

Scripts synthesise/prepare voice clips per language, merge audio layers, and maintain a manifest of every clip — making it straightforward to add or re-voice content.

Image toolchain

A prompt library plus a resumable render driver generate images locally; a compression step standardises every picture to the WebP target.

Content as data

Domains, items, and lessons are defined in structured data files, so new vocabulary drops in without touching the engine.

Reproducible builds

Media and manifests are version-controlled, so any release can be rebuilt and audited — important for a fundable, maintainable programme.

11.10 • Engineering principles

- **Offline-first, not offline-tolerant.** The network is used only for optional install/updates — never to deliver a lesson.
- **Single-artifact simplicity.** A self-contained package a facilitator can side-load and run, minimising IT overhead at the centre.
- **Cross-platform by default.** Open web technology runs on the widest range of low-cost devices without per-platform rebuilds.
- **Swappable content & language layers.** Audio, images, and vocabulary are data — the path to new languages without re-engineering.
- **Privacy by construction.** The architecture collects essentially nothing; the speech pipeline never leaves the device.

BUILD STATUS

The core enabling technology — real speech recognition running entirely offline on low-cost hardware — is implemented and working in the current prototype. The engineering scope funded by this grant is *hardening* (installable offline packaging, provisioning, facilitator reporting) and *content*, not unproven core technology.

12 Privacy, Safety & Data

Child-safe by construction — the safest data is the data never collected.

Voice never leaves the device

Speech recognition runs entirely on-device. No child's voice recording is transmitted, stored remotely, or sent to any third party.

No accounts, no personal data

Profiles use a first name or avatar only. No email, phone, location, or identifying data is required or collected.

No ads, no tracking, no in-app purchases

There is no advertising, behavioural tracking, or commercial content of any kind directed at children.

Local, exportable progress

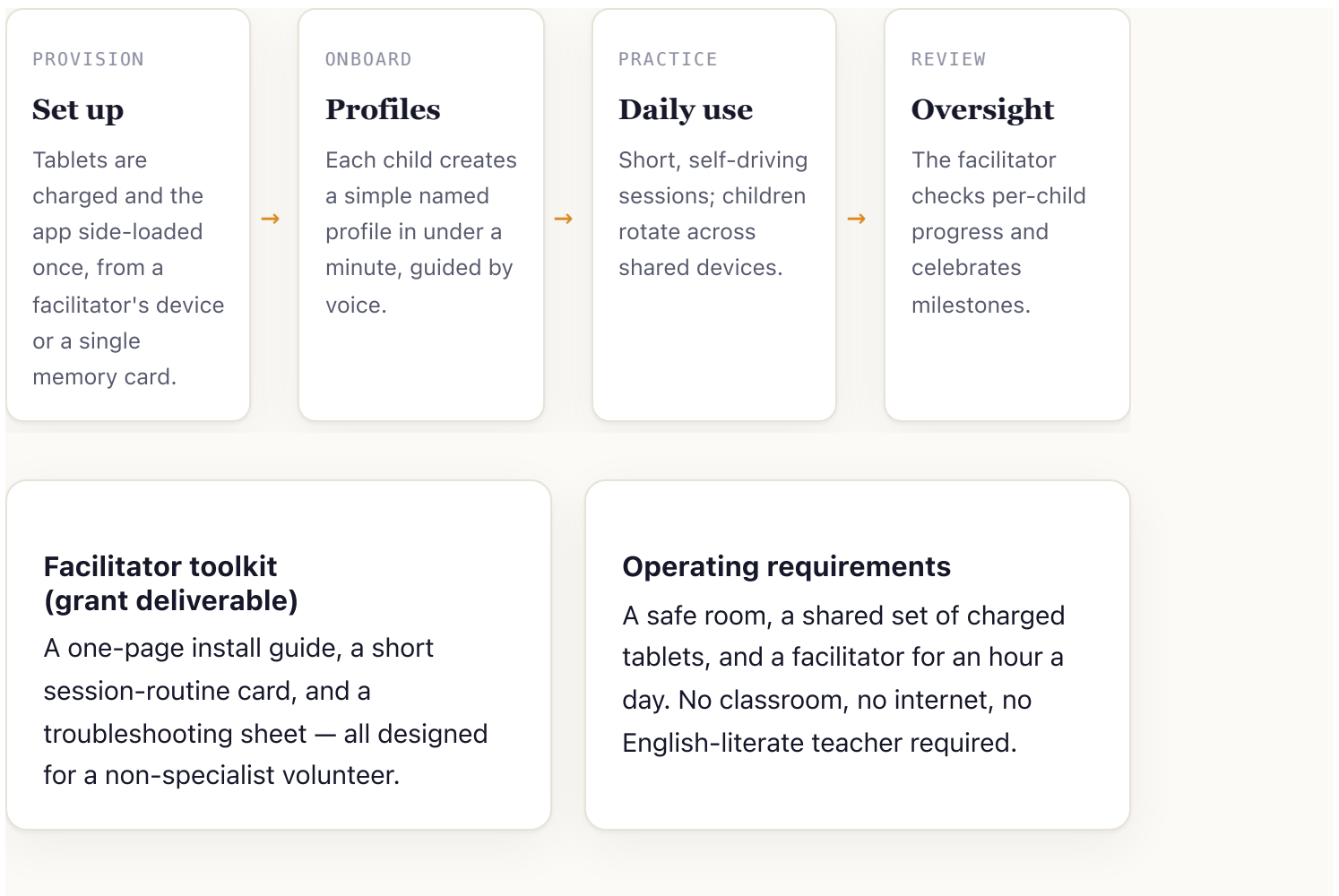
Progress data stays on the device and can be summarised for the facilitator; aggregate, de-identified metrics support evaluation.

ALIGNMENT

The design is built to respect children's-data and privacy principles (e.g. data minimisation and purpose limitation) by collecting essentially nothing — the final programme will complete a safeguarding and data-protection review with the host organisation before pilot.

13 Deployment & Operations

The learning-centre model — how Kratu actually reaches children.



14 Roadmap & Milestones

The grant period, phased from prototype to evaluated pilot. Durations are indicative.

PHASE 0 · DONE Prototype - Offline speech working - A–Z + 13 domains - Spelling & reading - Laila voice guide	PHASE 1 · Q1–Q2 Content & polish - Curriculum completion - Vocabulary depth - Native-speaker review - Accessibility pass	PHASE 2 · Q2–Q3 Field-ready build - Hardened installer - Facilitator toolkit - Device provisioning - Progress reporting	PHASE 3 · Q3–Q4 Monitored pilot - Centre deployment - Baseline assessment - Supervised daily use - Data collection
PHASE 4 · Q4 Evaluate & share - Endline assessment - Independent analysis - Learnings report - Scale-up plan			

Timeline assumes a ~12-month grant period. Replace quarters with grant-specific dates before submission.

15 Monitoring & Evaluation

How we will know Kratu works — and prove it to funders.

#	INDICATOR	METHOD	TARGET
M1	Letter recognition gain	Pre/post one-to-one assessment	[e.g. +X letters]
M2	Spoken-word vocabulary gain	Pre/post picture-naming task	[e.g. +X words]
M3	Early decoding / reading	Word-reading probe	[% reading N words]
M4	Engagement / dosage	On-device usage summaries	[min/child/week]
M5	Retention over the pilot	Active-learner tracking	[% retained]
M6	Facilitator usability	Structured interview / survey	[qualitative]

Approach

- **Baseline & endline.** Each child is assessed on entry and again after the pilot, with the difference as the primary learning measure.
- **Built-in dosage data.** The app's own usage summaries provide objective practice-time evidence without extra instrumentation or any personal data.
- **Mixed methods.** Quantitative gains are paired with facilitator and (where appropriate) caregiver interviews for the "why" behind the numbers.
- **Independent analysis.** Evaluation is designed with, and reviewed by, an external partner to ensure credibility.

16 Impact & SDG Alignment

SDG 4 — Quality Education

Directly advances foundational literacy (target 4.6) and early-childhood learning readiness (4.2) for excluded children.

SDG 5 — Gender Equality

Reaches girls who are disproportionately excluded from schooling, in a private, safe, at-the-centre setting.

SDG 10 — Reduced Inequalities

Targets exactly the rural, low-connectivity, mother-tongue communities that mainstream edtech bypasses.

Who benefits, and how far it reaches

[N]

Children directly reached in the pilot

[N]

Devices deployed across [N] centres

1:many

Each device serves many children on rotation

∞

Marginal cost of an extra lesson, once installed

COST CHARACTERISTICS

Because delivery is offline and self-driving, the marginal cost of teaching one more child one more lesson is effectively zero. Reach is bounded by devices and centres rather than by teachers or bandwidth.

17 Risks & Mitigations

Principal risks, positioned by likelihood and impact, each with a mitigation.

	LOW IMPACT	MED IMPACT	HIGH IMPACT
HIGH LIKEL.	R4 Device sharing limits practice time	R1 Speech accuracy across child voices/accents	R2 Facilitator capacity / turnover
MED LIKEL.	Content review backlog	R5 Hardware / power reliability	R3 Pilot recruitment & retention
LOW LIKEL.	Cosmetic UI issues	Storage constraints	R6 Safeguarding / data concerns

#	RISK	MITIGATION
R1	Recogniser under-performs on some child voices or accents.	Grammar-constrained matching, forgiving retries, native-speaker tuning, and continuous accuracy monitoring in the pilot.
R2	Facilitators are stretched or turn over.	Self-driving lessons minimise facilitator load; a one-page toolkit makes any volunteer replaceable quickly.
R3	Pilot recruitment or retention falls short.	Partner with an established community/host organisation; buffer cohort size; track and act on retention weekly.
R4	Shared devices cap per-child practice time.	Fast profile switching, short sessions, and device-to-child ratio planning at each centre.
R5	Power or hardware unreliability.	Battery/solar operation, rugged low-cost devices, and spares in the deployment budget.
R6	Safeguarding or data-protection concerns.	Collect essentially no personal data; on-device only; complete a safeguarding review with the host before pilot.

18 Sustainability & Scaling

Low marginal cost

Once built and installed, additional lessons and additional children cost effectively nothing — the economics improve with scale.

Replicable template

Content and language are data layers. A new language or region reuses the entire engine — the biggest cost is content, not code.

Partner-delivered

Delivery rides on existing community, NGO, and faith-based networks rather than requiring new institutions.

Open, durable technology

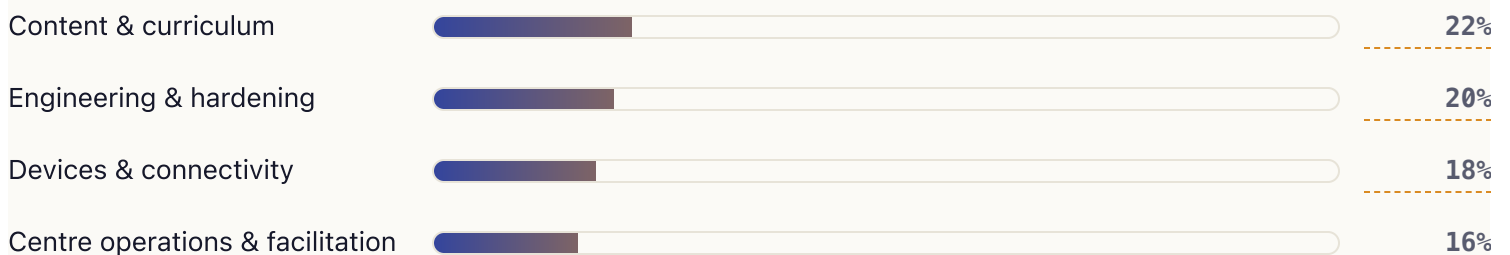
Built on open web standards with no proprietary lock-in, keeping long-term maintenance affordable.

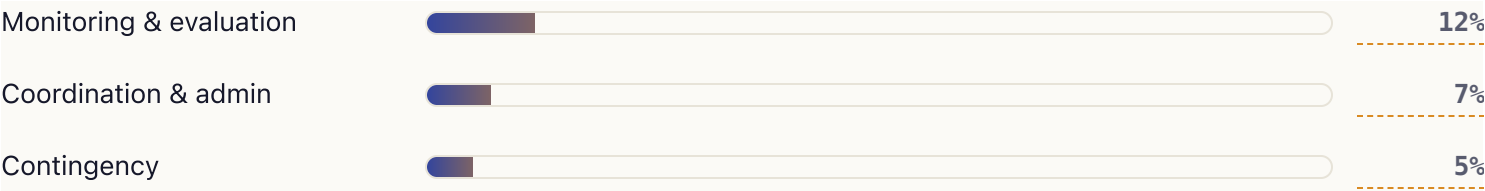
Beyond the grant

- **Prove it once, template it many times.** A successful, evaluated pilot becomes the reference model for replication across regions and languages.
- **Diversify funding.** Pilot evidence unlocks follow-on philanthropic, institutional, and government channels for scale-up.
- **Grow the content library** collaboratively with local educators, compounding value for every future deployment.

19 Budget Framework

Indicative allocation across categories. Replace percentages and amounts with your grant-specific figures.





CATEGORY	WHAT IT COVERS	AMOUNT
Content & curriculum	Vocabulary expansion, audio production, native-speaker review, educator design.	[amount]
Engineering & hardening	Installer, provisioning, progress reporting, accessibility, QA.	[amount]
Devices & connectivity	Tablets, charging/solar, spares, optional satellite for updates.	[amount]
Centre operations	Facilitator stipends, space, training, materials.	[amount]
Monitoring & evaluation	Assessment tools, enumerators, external analysis, reporting.	[amount]
Coordination & admin	Project management, safeguarding, financial administration.	[amount]
Contingency	Buffer for hardware, exchange, and field surprises.	[amount]
Total request	—	[total]

20

Team & Governance

Roles required to deliver. Complete with named individuals and partners before submission.

ROLE	RESPONSIBILITY	HELD BY
Project lead	Overall delivery, budget, funder reporting.	[name]
Product & engineering	App development, offline speech, releases.	[name]
Learning design	Curriculum, pedagogy, assessment design.	[name / partner]
Hausa language & content	Translation, voice, cultural & pronunciation review.	[name / partner]
Field & community partner	Centre access, facilitation, safeguarding.	[org]
Monitoring & evaluation	Baseline/endline, independent analysis.	[partner]

GOVERNANCE

A lightweight steering check at each phase gate (§14) reviews progress, spend, safeguarding, and evaluation integrity against this document.

21 Requirements Register

Functional and non-functional requirements, prioritised. Must Should Could Built

Functional requirements

ID	REQUIREMENT	PRIORITY
FR-1	All navigation and instruction is operable by voice + touch, with no reading required.	Built
FR-2	Recognise a child's spoken word/letter on-device and give immediate correct/retry feedback.	Built
FR-3	Deliver staged alphabet learning (taught groups → mixed → test).	Built
FR-4	Deliver vocabulary learning across all domains (hear → identify → speak → verify).	Built
FR-5	Deliver spelling (voice, letter-by-letter) and reading (whole-word) modes.	Built
FR-6	Maintain per-child profiles and progress on shared devices.	Built
FR-7	Match instruction grammar to the child's gender in Hausa.	Built
FR-8	Provide a facilitator progress summary per child.	Should
FR-9	Support early phrase/sentence practice beyond single words.	Could

Non-functional requirements

ID	REQUIREMENT	PRIORITY
NFR-1	Fully functional with zero network connectivity after install.	Must
NFR-2	Runs on entry-level tablets within typical storage and memory limits.	Must
NFR-3	No child audio or personal data leaves the device.	Must
NFR-4	Installable by a non-technical facilitator via a one-page guide.	Should
NFR-5	Content & language layers swappable without engineering changes.	Should
NFR-6	Operable on battery/solar with modest power draw.	Should
NFR-7	Respects reduced-motion and calm, low-stimulation design.	Could

22 Appendices & Sources

A • Glossary (selected Hausa terms)

- **Kratu** — the application / learning programme
- **Laila** — the on-screen voice guide (mascot)
- **Allo** — writing board / slate (the "back" motif)
- **Haruffa** — letters / alphabet
- **Abubuwa** — things / objects
- **Rubutu** — writing / spelling
- **Karatu** — reading
- **Launuka** — colours
- **Siffofi** — shapes
- **Lambobi** — numbers
- **Dabbobi** — animals
- **Abinci** — food

B • Suggested sources to cite (replace ‡ figures)

- UNESCO Institute for Statistics (UIS) — out-of-school children & literacy data.
- UNICEF Nigeria — education and out-of-school children briefings.
- Nigeria National Bureau of Statistics (NBS) — literacy & education surveys.
- Nigeria Demographic and Health Survey (NDHS) — regional/gender literacy indicators.
- Federal Ministry of Education / UBEC — national education statistics.
- Ethnologue / language references — Hausa speaker population.

BEFORE YOU SUBMIT

Replace every amber placeholder and every ‡-marked figure with grant-specific, cited values. Confirm the timeline, budget total, team, partners, and pilot size with your host organisation.

C • Current build status

A working single-file prototype exists with: staged A–Z lessons; thirteen vocabulary domains; spelling and reading modes; the Laila voice guide with gender-aware instruction; on-device speech recognition; per-child progress; and a photorealistic, culturally-specific image library — all running fully offline.

KRATU • PRODUCT REQUIREMENTS DOCUMENT • v1.0

Prepared to accompany a grant application. All amber placeholders and ‡-marked statistics are to be completed with verified, grant-specific figures before submission. This document describes intended outcomes; learning gains are to be measured in the pilot described in §15.